

Exercise 2

SQL Aggregates & operators

(1) Table Students

Student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

(1) SELECT DISTINCT department
FROM Students

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg-age
FROM Students

Group B) department	avg-age	department	avg-age
		Finance	23
		HR	22
		IT	20

3. SELECT department,
COUNT(*) AS student-count

FROM Students

Group B4 department

HAVING COUNT(*) > 1;

department	student-count
IT	2
HR	2

4. SELECT student_id, name
age, department
FROM Students
WHERE age BETWEEN 21 AND 23

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	2	HR

5. SELECT student_id, name
age, department
FROM Students
WHERE department IN 'IT' OR 'HR'
AND age > 21

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

TABLE Courses

Course_id	Course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6) SELECT department
 SUM(credits) AS total_credit
 FROM courses
 GROUP BY department
 HAVING SUM(credits) > 5

department	total_credits
IT	11

7) SELECT course_id, course_name
 department, credits
 FROM courses
 WHERE credits <> 4

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8) SELECT course_id, course_name
 credits
 FROM courses
 ORDER BY credits DESC
 LIMIT 3

course_id	course_name	credits
102	IT	4
103	IT	4
101	HR	3

9. en Table enrolments

enrolme_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

9. SELECT grade

MAX(grade) AS maximum_grade,

MIN(grade) AS minimum_grade,

AVG(grade) AS Avg_grade

FROM enrolments

Maximum_grade	Minimum_grad	Avg_grade
90	78	84.6

10. SELECT course_id,

COUNT(course_id) AS enrolment_count

FROM enrolments

GROUP BY course_id;

Table salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	55000
5	Buteh	HR	54000	3500

11. SELECT department,

SUM(salary) AS total_salary

SUM(bonus) AS total_bonus

FROM salaries

GROUP BY department;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. ~~SELECT~~ department.
~~FROM salary~~
~~WHERE~~ ~~AVG~~(salary) > 55000

12. SELECT department,
 AVG(salary) AS avg_salary
 FROM salaries
 GROUP BY department
 HAVING AVG(salary) > 55000;

department	avg_salary
IT	61000
Finance	70000

13. SELECT employee_id,
 name, salary,
 bonus
~~FROM salary~~ Salary + bonus AS total_compensation
 FROM employees
 WHERE Salary + bonus > 60000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
2	Spike	70000	6000	76000
3	Tyke	62000	5500	67500

TABLE Projects

Project_id	project_name	department	budget
1	AI App	IT	12000
2	Payroll system	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

SELECT department,
 SUM(budget) AS total_budget,
 AVG(budget) AS avg_budget

FROM projects

GROUP BY department

HAVING AVG(budget) > 10000;

department	total_budget	avg_budget
11	270000	135000
Finance	80000	80000

is. SELECT project_id,
 project_name,
 department,
 budget

FROM Projects

WHERE budget BETWEEN 50000 AND 120000

AND department <> 'Marketing';

project_id	project_name	department	budget
1	AI App	11	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000